Experiment 3

Emerging Trends in Low Code Application Development

1. Aim

To study the emerging trends in low-code application development and to analyse how each trend applies to the Gym Membership Management application built on Zoho Creator.

2. Introduction

Low-code platforms began as visual tools for building simple forms-and-database applications. They have since grown into broader application platforms that include automation, integration, analytics and artificial intelligence. Vendors such as Zoho, Microsoft (Power Apps), OutSystems and Mendix compete in this space, and the trends below are visible across them.

This record is a literature-based study. It describes each trend, explains why it matters, and links it to the gym domain.

3. Procedure

- Reviewed official documentation and product pages of leading low-code platforms.
- Identified recurring trends across vendors.
- Described each trend and its benefits and risks.
- Mapped each trend to a feature or future improvement of the Gym application.

4. Emerging Trends

Trend	Description	Application to the Gym app
AI-assisted development	Platforms add AI to suggest fields, generate apps or scripts from plain-language prompts, and answer questions about data	Describe the gym app in words and generate a starting set of forms; ask questions about attendance data
Citizen development	Business users, not only programmers, build and change applications	Gym manager adjusts plans and reports without waiting for a developer
Hyperautomation	Combining workflows, integrations, AI and robotic process automation to automate whole processes	Renewal reminders, payment follow-up and receipt generation without manual work
API-first and integration	Platforms connect easily to other services through REST APIs and prebuilt connectors	Link to Zoho Books for invoices, email and messaging for alerts, payment gateways
Pro-code extensibility	Developers extend low-code apps with custom functions, scripts and external services	Custom late-fee calculator or an external biometric entry service
Multi-experience apps	One application delivered to web, mobile and other channels	Members check plans and attendance on a phone; staff use the web
Governance and security	Central control of who can build, change and deploy apps, plus audit trails	Role-based access, audit of fee changes, controlled deployment from development to production
Embedded analytics	Dashboards and insights built into the platform	Dashboard of active members, expiries and revenue trends

Trend	Description	Application to the Gym app
Composable and reusable components	Applications assembled from reusable pieces and templates	Reuse the Member and Payment forms in a second branch app

5. Benefits and Risks

Aspect	Benefit	Risk
Speed	Applications reach users quickly	Quick builds may skip design and testing
Accessibility	More people can build solutions	Uncontrolled apps (shadow IT) and inconsistent data
AI features	Less manual work, faster starts	Generated output needs review; accuracy and data privacy must be checked
Integration	Existing tools stay in use	Dependency on external services and their limits
Vendor platform	No infrastructure to manage	Vendor lock-in and changing pricing or features

6. Discussion

For a gym business the most relevant trends are hyperautomation, integration and embedded analytics, because the daily work is repetitive: renewals, reminders, payments and attendance. AI assistance is promising for quick prototyping and for natural-language questions about member data, but its results should be reviewed before use, especially for payment and personal data.

Governance becomes important as soon as several staff members use the system. Role-based access, audit records and a separate production version protect member data and prevent accidental changes. Pro-code extensibility reduces the main weakness of low-code, since unusual rules can be handled in custom functions while the rest of the app stays visual.

Overall, the trends point to low-code platforms becoming more intelligent, more connected and more tightly governed, which supports using Zoho Creator for this project.

7. Result

Nine emerging trends in low-code development were identified and mapped to the Gym Membership Management application. Hyperautomation, integration, embedded analytics and governance were found to be the most directly useful for this project.

Evidence boundary: this is a literature-based study of vendor documentation and general sources. It reports trends qualitatively and does not include measured data or live platform tests.

8. References

- Zoho Creator: https://www.zoho.com/creator/
- Zoho Creator help: https://www.zoho.com/creator/help/

- Microsoft Power Apps: https://powerapps.microsoft.com/
- OutSystems: https://www.outsystems.com/
- Mendix: https://www.mendix.com/
- Low-code development platform (overview): https://en.wikipedia.org/wiki/Low-code_development_platform